European Journal of Endocrinology, 2023, **189**, K1–K6
https://doi.org/10.1093/ejendo/lvad093
Advance access publication 20 July 2023
Brief Report

Etiology, morphology, and outcomes of adrenal calcifications in 540 adult patients—a retrospective single-center study

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Abstract

Sparse data are available on the etiology, morphology, clinical presentation, and outcomes of adrenal calcification. In this single-center retrospective study of 540 consecutive patients (median age 65 years, 45% women) with adrenal calcification on computed tomography (December 2017 to January 2021), most were discovered incidentally (472, 87%). The commonest etiology was idiopathic (389, 72%), followed by the adrenal tumor (113, 21%), hemorrhage (29, 5%), and infiltrative disease (7, 1%). Calcified adrenal tumors were predominantly benign (92, 81%) and primarily adenomas (63, 69%), whereas the most common calcified adrenal malignancy (16, 18%) was metastasis (12, 67%). Calcification (unilateral 94%, bilateral 6%) morphology varied from punctate (313, 58%) to coarse (165, 30%), linear/curvilinear (46, 9%), and rim-like (16, 3%). In summary, adrenal calcifications are usually incidentally discovered unilateral, punctate, or coarse lesions of unclear etiology. Most calcified adrenal tumors are benign and <20% malignant or pheochromocytomas.

Keywords: computed tomography, calcified, adrenal tumor, primary adrenal insufficiency, adrenal mass

Significance

Existing literature on adrenal calcification is mainly limited to case reports. We evaluated a large cohort of adult patients with adrenal calcification and found that most were idiopathic. A fifth of all adrenal calcifications were associated with an underlying adrenal tumor. Most calcified adrenal tumors were benign and predominantly adenomas, whereas metastasis represented the most common malignant etiology. No specific calcification morphology was pathognomonic of a malignant adrenal tumor. The findings of this study, for the first time in the literature, provide an insight into the clinical presentation, various etiological associations, and morphology of adrenal calcifications.

Introduction

The deposition of calcium in blood vessels, organs, and tumors may reflect an underlying pathology, disease severity, or chronicity.[1–3] Adrenal calcification may be a result of a previous adrenal hemorrhage, infection, and inflammation or occur within an adrenal tumor.[4–9] Our understanding of adrenal calcification remains poor, with published literature limited to case reports on primarily infants and children, several review articles, and 2 small case series of patients with calcified adrenal masses, including 1 that was published more than 3 decades ago.[5–9] Keeping in mind the knowledge gaps, the paucity of current literature, and the increased radiological incidence of adrenal calcification, we designed our current study to determine the etiological associations and describe the clinical presentation, morphology, and outcomes of adrenal calcification.

Methods

Study design and patient selection

This is a retrospective cohort study that was approved by the Institutional Review Board of Mayo Clinic and complies with the Declaration of Helsinki. A research consent waiver was used. We employed 2 search strategies to identify adult patients with adrenal calcification between January 1, 2017, and December 31, 2021. First, we electronically interrogated our institutional radiology database for patients who underwent computed tomography (CT) imaging of the chest, abdomen, pelvis, and thoracic or lumbar spine over the 5-year study period using the search terms "adrenal" combined with "calcification" or "calcified." This search revealed 471 (0.14%) patients out of the 342 298 unique patients with CT imaging performed. Next, we used our longitudinal Adrenal Mass Database, which has been previously utilized in several published studies and comprises 8032 consecutive patients with adrenal masses (including 678 patients with

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Received: February 28, 2023. **Revised:** April 23, 2023. **Editorial Decision:** May 22, 2023. **Accepted:** May 22, 2023
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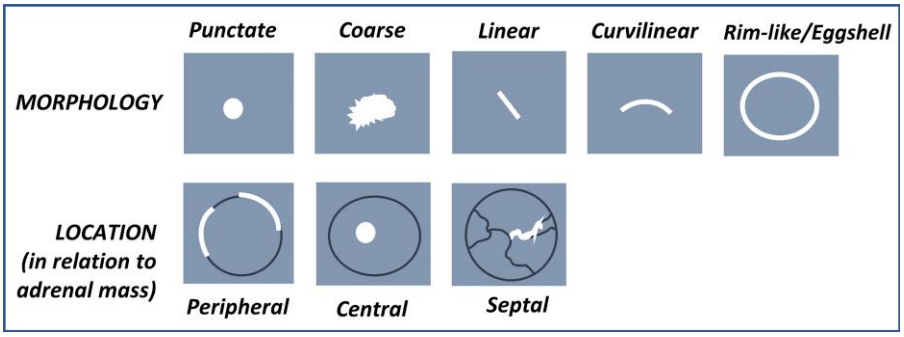


Figure 1. Schematic presentation of adrenal calcification morphologies and location.

pheochromocytoma, 598 with adrenal metastasis, and 213 with adrenocortical carcinoma) evaluated at our institution between January 1, 1995, and December 31, 2021.[10–15] The electronic search of the Adrenal Mass Database resulted in 571 potential patients with a calcified adrenal tumor (after excluding 248 patients that overlapped with the radiology database) during the study period. The search results of the 2 databases were combined, and radiology reports and associated CT images were reviewed. Those with unavailable CT images or lack of calcification associated with adrenal gland(s) or tumors were excluded.

Data collection

Medical records of patients in the final cohort were reviewed, and information was gathered on demographics, mode of initial discovery, etiology, imaging characteristics, and outcomes. Computed tomography images were analyzed for the adrenal calcification laterality, number, morphology (punctate, coarse, linear/curvilinear, and rim-like/eggshell), and association with an adrenal mass and location (peripheral, septal, central, and combination) (Figure 1). As described in the literature, calcification was identified as bright white spots and unenhanced attenuation of ≥100 Hounsfield units on CT imaging.[3] Outcomes of interest were etiology and if associated with an adrenal mass—malignancy. Malignancy was diagnosed based on histopathology. For calcifications without evidence of an apparent underlying cause, etiology was labeled as "idiopathic." Subgroup comparisons were made based on the calcification laterality (unilateral vs bilateral), association with an underlying adrenal tumor (with vs without adrenal tumors), and nature of the adrenal tumor (benign vs malignant).

Statistical analysis

We presented categorical data as counts and percentages and continuous data as median and interquartile ranges (IQR). The association between variables was assessed using the chi-square test and Wilcoxon/Kruskal–Wallis test. Statistical significance was defined as a *P*-value of <.05. Statistical analyses were conducted using the JMP Software, version 16.1.0, SAS Institute Inc.

Results

Over the 5-year study duration, radiological evidence of adrenal calcification was noted in 540 patients at a median age of 65 years (IQR, 54-74), comprising 245 (45%) women and 512 (95%) White individuals. Adrenal calcification was discovered incidentally on CT imaging performed for unrelated reasons in 472 (87%) patients, during cancer staging imaging in 66 (12%), and for other reasons in 2 (1%). Only 167 (31%) patients were evaluated by an endocrinologist. The etiology of adrenal calcification was idiopathic in 389 (72%), related to an underlying adrenal tumor in 113 (21%), adrenal hemorrhage in 29 (5%), and infiltrative disease (infectious or inflammatory) in 7 (1%) patients. Of the 113 patients with adrenal tumors, benign tumors were diagnosed in 92 (81%), adrenal malignancy in 18 (16%), and pheochromocytomas in 2 (2%), and 1 (1%) patient had an inflammatory pseudotumor. Among the 92 benign adrenal tumors, most were adrenal adenomas (63, 69%), and for the 18 malignant tumors, metastases (12, 67%) were the most common. Overall, only 7 (1%) patients had associated primary adrenal insufficiency, diagnosed before the appearance of calcification in all, and 18 (3%) underwent adrenalectomy, primarily due to concern for underlying malignancy (Table 1).

Adrenal calcification(s) were unilateral in 506 (94%) patients (right 264, 49%; left 242, 45%) and bilateral in 34 (6%), with the majority presenting as a single calcification (415, 77%). The most common calcification morphology was punctate (313, 58%), followed by coarse (165, 30%), linear or curvilinear (46, 9%), and rim-like (16, 3%). When associated with an adrenal tumor, the pattern of calcification was central in 51 (45%), peripheral in 43 (38%), septal in 10 (9%), and multiple locations in 9 (8%) patients (Figure 2, Table 1).

Compared to patients with unilateral adrenal calcification, those with bilateral involvement were more likely to be discovered during cancer staging (18% vs 12%, $P = .02$), have associated primary adrenal insufficiency (12% vs 1%, $P < .0001$), present as multiple (53% vs 21%, $P < .0001$) and coarse lesions (62% vs 28%, $P < .0001$), and to be associated with an adrenal tumor in multiple locations (40% vs 5%, $P = .001$) (Table 1). Patients with calcified adrenal tumors, compared to those with non-neoplastic calcification, were slightly younger with a higher proportion of women (Table 2). Subgroup analysis of malignant vs benign tumors revealed no difference in the calcification morphology, but those with malignant tumors were less likely to present with calcifications in peripheral (11% vs 43%) or septal (5% vs 10%) locations ($P = .03$) (Table 2).

Discussion

In this largest-to-date study on adrenal calcification, we describe the clinical presentation, etiology, morphology, and

Table 1. Clinical characteristics and radiological presentation of adrenal calcification for the entire cohort and based on the laterality.

	All patients $N = 540$	Unilateral $N = 506$	Bilateral $N = 34$	*P*-value
Clinical characteristics				
Women, *n* (%)	245 (45)	233 (46)	12 (35)	.28
Age at diagnosis, years, median (interquartile range)	65 (54-74)	65 (54-74)	66 (52-77)	.95
Race, *n* (%)				
European American	512 (95)	480 (95)	32 (94)	.19
African American	5 (1)	5 (1)	0 (0)	
Asian American	7 (1)	7 (1)	0 (0)	
Other or mixed	16 (3)	14 (3)	2 (6)	
Mode of discovery, *n* (%)				
Incidental	472 (87)	445 (88)	27 (79)	.02
Cancer staging imaging	66 (12)	60 (12)	6 (18)	
Other	2 (1)	1 (<1)	1 (3)	
Etiology, *n* (%)				
Idiopathic	389 (72)	369 (73)	20 (59)	.24
Adrenal tumor	113 (21)	103 (21)	10 (29)	
Hemorrhage	29 (5)	25 (5)	4 (12)	
Infiltrative disease[a]	7 (1)	7 (1)	0 (0)	
Post-adrenalectomy	2 (<1)	2 (<1)	0 (0)	
Type of adrenal tumors, *n* (%)				
Adenoma	63 (56)	59 (57)	4 (40)	.10
Myelolipoma	13 (12)	9 (9)	4 (40)	
Benign adrenal cyst	15 (12)	14 (14)	0 (0)	
Pheochromocytoma	2 (2)	1 (1)	1 (10)	
Adrenocortical cancer	5 (4)	5 (5)	0 (0)	
Metastasis[b]	12 (11)	11 (11)	1 (10)	
Other[c]	4 (3)	4 (3)	0 (0)	
Evaluated by an endocrinologist, *n* (%)	167 (31)	153 (30)	14 (41)	.18
Primary adrenal insufficiency[d]**,** *n* (%)	7 (1)	3 (1)	4 (12)	<.0001
Adrenalectomy[e]**,** *n* (%)	18 (3)	17 (3)	1 (3)	.89
Imaging characteristics				
Site, *n* (%)				
Right	264 (49)	—	—	—
Left	242 (45)			
Bilateral	34 (6)			
Shape, *n* (%)				
Punctate	313 (58)	303 (60)	10 (29)	<.0001
Coarse	165 (30)	144 (28)	21 (62)	
Curvilinear or linear	46 (9)	43 (9)	3 (9)	
Rim-like (eggshell)	16 (3)	16 (3)	0 (0)	
Location in relation to the adrenal tumor[f]**,** *n* (%)				
Central	51 (45)	48 (47)	3 (30)	.001
Peripheral	43 (38)	40 (39)	3 (30)	
Septal	10 (9)	10 (10)	0 (0)	
Multiple locations	9 (8)	5 (5)	4 (40)	
Number of calcifications in each site, *n* (%)				
Single	415 (77)	399 (79)	16 (47)	<.0001
Multiple	125 (23)	107 (21)	18 (53)	

[a]Include histoplasmosis ($n = 1$), amyloidosis ($n = 1$), tuberculosis ($n = 1$), and unspecified granulomatous disease ($n = 4$).
[b]Primary malignancy—5 lung cancer, 2 breast cancer, 2 renal cancer, 1 testicular cancer, 1 cholangiocarcinoma, 1 melanoma, and 1 thoracic synovial sarcoma.
[c]Include adrenal plasmacytoma ($n = 1$), teratoma ($n = 1$), adenomatoid tumor ($n = 1$), and inflammatory pseudotumor ($n = 1$).
[d]Etiology of primary adrenal insufficiency in patients with bilateral calcification was bilateral adrenal hemorrhage ($n = 3$) and bilateral adrenal metastases ($n = 1$), and in patients with unilateral calcification: adrenomyeloneuropathy ($n = 1$), histoplasmosis and hemorrhage ($n = 1$), and bilateral adrenal hemorrhage ($n = 1$).
[e]Histopathology—adrenocortical carcinoma ($n = 4$), metastasis ($n = 3$), pheochromocytoma ($n = 1$), teratoma ($n = 1$), adenoma ($n = 3$), benign adrenal cyst ($n = 2$), myelolipoma ($n = 2$), adenomatoid tumor ($n = 1$), and inflammatory pseudotumor ($n = 1$).
[f]Recorded for 113 patients with calcification associated with neoplasm (10 bilateral neoplasms and 103 unilateral neoplasms).

outcomes in a consecutive cohort of adult patients. We found that almost all adrenal calcifications were discovered on imaging performed for evaluation of a non-adrenal pathology and usually presented as single, unilateral, punctate, or coarse lesions of unclear etiology. We also found that most adrenal tumors demonstrating calcification were benign; however, adrenal malignancy and pheochromocytomas represented 18% of patients with adrenal tumors.

Most patients in our cohort were between 50 and 70 years, likely explained by a higher prevalence of comorbidities prompting the need for radiological investigations with advancing age. Despite the increased use of radiological examinations, we found adrenal calcifications to be rare, reported on 0.14% of abdominal CT scans. In our previous population-based study of patients with adrenal lesions, calcifications were reported in 0.5% of patients.[4]

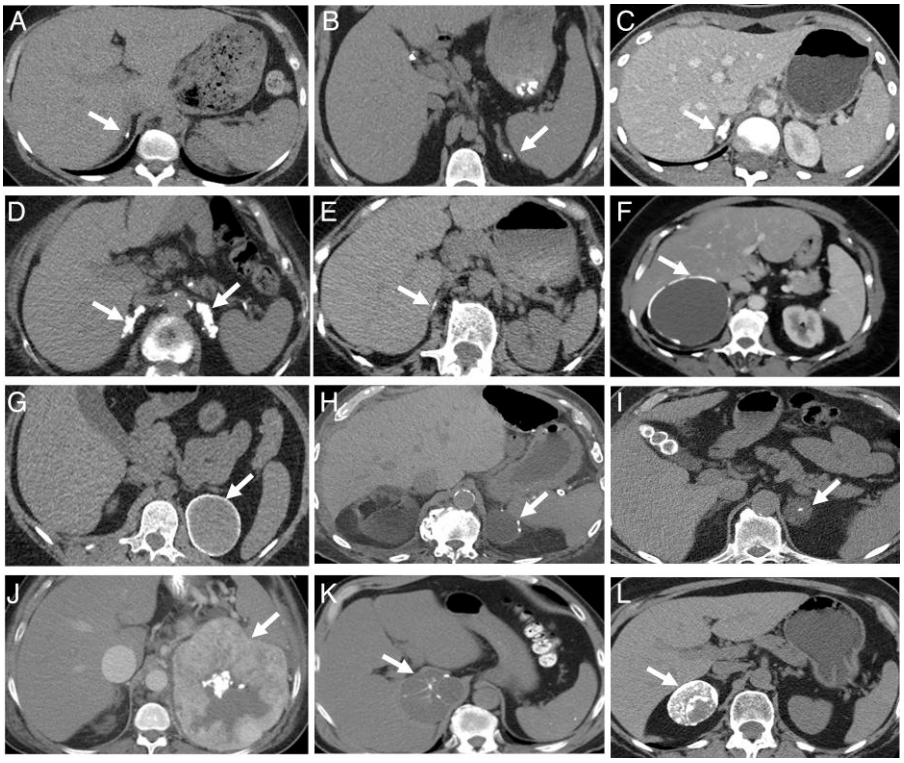


Figure 2. Adrenal (gland and tumor) calcification illustration on computed tomography imaging. (A) Punctate calcification in the right adrenal gland. (B) Punctate calcifications in the left adrenal gland. (C) Coarse calcification in the right adrenal gland. (D) Coarse calcifications of the bilateral adrenal glands. (E) Linear calcification in the right adrenal gland. (F) Curvilinear peripheral calcifications associated with the right adrenal cyst. (G) Rim-like (eggshell) calcification associated with the left adrenal cyst. (H) Punctate peripheral calcifications associated with the left adrenal adenoma. (I) Punctate central calcifications associated with the left adrenal adenoma. (J) Coarse central calcification associated with the left adrenal pheochromocytoma. (K) Punctate and linear septal calcifications associated with the right adrenal teratoma. (L) Coarse central and curvilinear peripheral calcification associated with the right adrenal cyst.

Table 2. Subgroup analysis of adrenal calcification based on association with an underlying adrenal tumor and nature of the adrenal tumor.

N	Without adrenal tumors 427	With adrenal tumors[a] 113	*P*-value	Benign adrenal tumor 92	Malignant adrenal tumor 18	*P*-value
Clinical characteristics						
Women, *n* (%)	176 (41)	69 (61)	.0002	58 (63)	10 (56)	.55
Age at diagnosis, years, median (interquartile range)	66 (54-75)	63 (52-70)	.01	63 (52-70)	57 (52-67)	.40
Mode of discovery, *n* (%)						
Incidental	381 (89)	91 (80)	.002	82 (89)	7 (39)	<.0001
Cancer staging	44 (10)	22 (20)		10 (11)	11 (61)	
Other	2 (1)	0 (0)		0 (0)	0 (0)	
Primary adrenal insufficiency, *n* (%)	6 (1)	1 (1)	.66	0 (0)	1 (6)	—
Adrenalectomy, *n* (%)	0 (0)	18 (16)	<.0001	9 (10)	7 (39)	.001
Imaging characteristics						
Site, *n* (%)						
Unilateral	403 (94)	103 (91)	.21	84 (91)	17 (94)	.66
Bilateral	24 (6)	10 (9)		8 (9)	1 (6)	
Shape, *n* (%)						
Punctate	256 (60)	57 (50)	.01	48 (52)	8 (44)	.26
Coarse	129 (30)	36 (32)		25 (27)	9 (50)	
Curvilinear or linear	34 (8)	12 (11)		11 (12)	1 (6)	
Rim-like or eggshell	8 (2)	8 (7)		8 (9)	0 (0)	
Number of calcifications, *n* (%)						
Single	354 (83)	61 (54)	<.0001	51 (55)	9 (50)	.67
Multiple	73 (17)	52 (46)		41 (45)	9 (50)	
Location in relation to the adrenal neoplasm, *n* (%)						
Central	—	51 (45)	—	37 (40)	12 (67)	.03
Peripheral		43 (38)		40 (43)	2 (11)	
Septal		10 (9)		9 (10)	1 (5)	
Multiple locations		9 (8)		6 (7)	3 (17)	

[a]Include 92 benign, 18 malignant, and 3 other (2 pheochromocytomas and 1 inflammatory pseudotumor).

Existing literature on adrenal calcification is sparse. The 2 prior studies focused only on patients with calcified adrenal tumors, 1 being a surgical series and another published more than 3 decades ago describing mainly non-incidental cases, thereby limiting the evaluation of other etiological associations.[7,9] In contrast, in our study, we used broader inclusion criteria and identified patients from 2 separate institutional databases. Though we found that most adrenal calcifications were idiopathic, approximately 30% had an identifiable etiology, with an underlying adrenal tumor being the most common, followed by hemorrhage and, rarely, a benign infiltrative disease. Notably, most calcified adrenal tumors were benign and predominantly adenomas but also included myelolipomas and adrenal cysts. For the minority with underlying adrenal malignancy, most were adrenal metastases, and less than one-fourth represented either adrenocortical cancer or pheochromocytoma. In a recent study by Yalon et al.,[16] where they defined and evaluated a novel 9-variable image-based score to differentiate adrenocortical carcinomas and lipid-poor adrenal adenomas, calcification was negatively predictive of adrenocortical carcinoma. Despite the differences in the study design and cohort selection, these findings align with our observations that most calcified adrenal tumors are benign. In our study, we reviewed imaging characteristics in all patients and appreciated various morphologies of adrenal calcifications. While we found no specific calcification morphology that distinguished malignant from benign adrenal tumors with 100% accuracy, we noted that the peripheral or septal location of calcification was less likely to be associated with malignant tumors.

In addition to the large sample size, the strengths of our study are the inclusion of consecutive patients irrespective of etiology or management, thereby enabling a better understanding of the prevalence, etiological associations, and outcomes in adrenal calcifications. Nonetheless, our study has certain limitations mainly related to the retrospective single-center design that led to information and selection bias and the short duration of our study. Our study's adult and predominantly White population from the northern mid-west region of the United States may have had an impact on the prevalence of certain etiologies and limit the generalizability of these findings to children, non-White races, and those in different geographical locations. As the diagnosis of calcification was based on the availability of CT, imaging bias is present in our assessment of the calcification prevalence. We also acknowledge the lack of histopathology in most of our cohort, including those with adrenal tumors; however, given the unique presentation on CT, misdiagnosis of calcification was unlikely. The rare occurrence of primary adrenal insufficiency in our study could be an underestimation, especially in patients with bilateral adrenal metastasis, as many were likely not evaluated for adrenal insufficiency.[15]

In conclusion, adrenal calcifications are incidentally discovered lesions with a radiological prevalence of 0.14%, typically presenting as unilateral, punctate, or coarse lesions of unclear etiology. Only a fifth of calcifications are associated with an underlying adrenal tumor. Most adrenal tumors with associated calcifications are benign, with less than 20% representing a pheochromocytoma or an adrenal malignancy. No specific calcification morphology helps distinguish malignant from benign adrenal tumors with 100% accuracy.

Funding

This research was partly supported by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and National Institute of Aging (NIA) of the National Institutes of Health (NIH) USA under awards K23DK121888, R03DK132121, and R03AG71934 (to I.B.). The views expressed are those of the authors and not necessarily those of the NIH USA.

Conflict of interest: I.B. serves on the editorial board for EJE and was not involved in the handling of this manuscript. I.B. reports consulting (fee to institution) from HRA Pharma, Recordati, Corcept, Sparrow, Neurocrine, Diurnal, Adrenas, and Spruce, outside this work. I.B. reports funding for the investigator-initiated award from Recordati (outside this work). Other authors declare no conflict of interest.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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